Homework 14: 10/27 Content

MATH 1271, Garrett Credi

10/27/2026

Problems listed as AC are found in the Active Calculus textbook at the end of the listed topic. You should be finding these in the *Exercises* section of the page.

Problems listed as APEX are found in the APEX Calculus Textbook again at the end of the listed topic. You should be finding these in the *Exercises* section of the page.

Problems listed as PAULS-PRACT are found in Paul's Online Math Notes - Review Problems while problems listed as PAULS-ASSIGN are found in Paul's Online Math Notes - Assignment Problems and are indexed via [Topic]-[Section]-[Problem Number]

1 Type 1 - L'Hôpital's Rule Analytically

Choose 8 of the following problems to solve.

1. AC 3.2.2
2. AC 3.2.3
3. AC 3.2.6-14 (each of these count as one problem)
4. APEX 6.7.9-54 (each of these counts as one problem) [hahaha 67]
5. PAULS-ASSIGN.Applications of Derivatives.L'Hospital's Rule and Indeterminate Form.1-18

2 Type 2 - L'Hôpital's Rule Graphically

Solve both of the problems listed below.

1. AC 3.2.4
2. AC 3.2.5

3 Challenge Problem

Find an example of two functions $f(x), g(x)$ so that:

1. $f(0) = g(0) = 0$
2. $\lim_{x \to 0} \dfrac{f(x)}{g(x)}$ exists
3. $\lim_{x \to 0} \dfrac{f'(x)}{g'(x)}$ does not exist.

(I promise that such a pair of functions does exist and they're not weird/esoteric).

Does this mean L'Hôpital's rule is false?